

readbackground

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17:15

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1 A program to read a background specification from a file

\$Id: 506bc1c6d96700f98a720d9a9c11d874f8672e93 \$

This program reads background plasma and neutral source data generated by some other code and translates it into the format needed by DEGAS 2. It is invoked with the command line

```
readbackground an_input_file
```

This code currently accepts three types of input files:

1. An input file for the original DEGAS code. The file specified on the command line, `an_input_file`, should end with `.d`.
2. A data transfer file generated by the UEDGE code. In this case, `an_input_file` should end in `.u`. This usage is consistent with the original implementation of *readbackground* in which only a single ion species is considered.
3. An input file pointing to a data transfer file generated by the UEDGE code and providing additional descriptive information. This option is selected by having `an_input_file` end in anything other than `.d` or `.u`.

The old DEGAS option has been used infrequently of late, although the DEGAS - DEGAS 2 benchmark remains in the DEGAS 2 examples. Consequently, newer DEGAS 2 features may not be available with this option.

The UEDGE data file is assumed to be in the format corresponding to the UEDGE subroutine *writemcnfile*. M. Rensink has used a BASIS script to create functionally equivalent files that will also work with this program. Presently, this program can only handle runs with two divertor targets, i.e., single null or symmetric double null geometries.

The third option indicated above provides compatibility with multispecies UEDGE runs and allows the user to specify plasma parameters for DEGAS 2 zones filling the volume between the UEDGE mesh and surrounding solid zones (e.g., created with *definegeometry2d*). This input file is of the sort used elsewhere in DEGAS 2. Namely, blank lines, spaces, and lines beginning with the comment character `#` are ignored; comments can also appear at the end of a line.

Each line in the input file is of the form:

keyword arguments

The available keywords are

uedge_file filename specifies the name of the UEDGE data file. In this case, *filename* does *not* need to end in `.u`.

ion_species **sp1 sp2 sp3 ...** provides the list of ion species appearing in the **UEDGE** run. The order of this list must correspond to the species ordering used by **UEDGE** in writing the file. The strings provided here should correspond to **DEGAS 2** species symbols (i.e., to entries in the *species_sy* array). The code will ignore data corresponding to unrecognized symbols; the code will generate a warning message in those cases. If a species is recognized, but not included in the problem input file for the current **DEGAS 2** run (e.g., an impurity species used in **UEDGE**, but not in **DEGAS 2**), its density will be used to compute the electron density via quasineutrality. Hence, to guarantee consistency of the electron density in **DEGAS 2** and **UEDGE**, all pertinent species should be added to the **DEGAS 2** reference species list. The flow velocity for such species will also be used in computing the total ion flow velocity that is used in the sheath potential expression. If a species in this list is recognized and is in the current problem input file, its density and flow velocity data will be read into the background density and velocity arrays. Its (nonzero) fluxes to the targets will be used to define source groups, one per species per target. These are always assumed to be plate (recycling) sources.

bdy_ni **sp** i_s n instructs the code to assign ion species **sp** (again, representing a species symbol; it must appear in the *ion_species* keyword and in the current problem input file) a density n (in m^{-3}) in zones associated with stratum i_s . This stratum number corresponds to those used in *definegeometry2d*'s *stratum* keyword. The stratum number there acts as a label for all zones created out of the polygon specified in conjunction with that keyword. Ensuring uniqueness of the stratum label is ultimately the user's responsibility, although the code may catch some duplications. The stratum labels are contained in the polygon netCDF file generated by *definegeometry2d*; the name of this file must be specified with the *polygon_file* keyword (see below).

bdy_vr **sp** i_s v_r instructs the code to assign ion species **sp** a radial flow velocity v_r (in m/s) in zones associated with stratum i_s . The cylindrical components of the flow velocity are provided; during **DEGAS 2** execution, they resulting vector is rotated to the toroidal angle where the information is needed.

bdy_vphi **sp** i_s v_ϕ instructs the code to assign ion species **sp** a radial flow velocity v_ϕ (in m/s) in zones associated with stratum i_s . The cylindrical components of the flow velocity are provided; during **DEGAS 2** execution, they resulting vector is rotated to the toroidal angle where the information is needed.

bdy_vz **sp** i_s v_z instructs the code to assign ion species **sp** a radial flow velocity v_z (in m/s) in zones associated with stratum i_s . The cylindrical components of the flow velocity are provided; during **DEGAS 2** execution, they resulting vector is rotated to the toroidal angle where the information is needed.

bdy_ti i_s T_i instructs the code to assign an ion temperature T_i (in eV) to all ion species in the problem (more specifically, all background species except electrons) in zones associated with stratum i_s .

bdy_te i_s T_e instructs the code to assign an electron temperature T_e (in eV) to zones associated with stratum i_s .

polygon_file **filename** specified the name of the polygon netCDF file generated *definegeometry2d*. The information in this file is needed to relate the stratum labels in the *bdy_ni*, etc. keywords to the zone numbers.

```
"readbackground.f" 1 ≡
  @m FILE 'readbackground.web'
```

The unnamed module.

"**readbackground.f**" 1.1 ≡

@f *namelist integer*

⟨ Functions and Subroutines 1.2 ⟩

The main program

⟨ Functions and Subroutines 1.2 ⟩ ≡

program *readbackground*

implicit_none_f77

implicit_none_f90

character**FILELEN* *datafilename*

call *command_arg*(1, *datafilename*)

call *process_background*(*datafilename*)

stop

end

See also sections 1.3 and 1.4.

This code is used in section 1.1.

The main subroutine.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine process_background(datafilename)
  implicit none
  zn_common
  pr_common
  bk_common
  so_common
  gi_common
  implicit none
  character*FILELEN datafilename, degasfilename, uedgefilename
  integer uedge, degas, update, geom_modified
@#if 0
  integer i, ii
@#endif

  ⟨ Memory allocation interface 0 ⟩

  call readfilenames

  degas = FALSE
  uedge = FALSE
  /* Find out whether the data file is DEGAS or UEDGE depending on the extension of the file
  name. Have now generalized the UEDGE case to permit some additional options or information
  to be specified in a text file. A DEGAS file is still denoted by a “.d” extension. To use just a
  UEDGE data file by itself, the user would have to give it a “.u” extension. */
  if (index(datafilename, '.d') > 0) then
    degasfilename = datafilename
    degas = TRUE
@#if 1
  else
    uedgefilename = datafilename
    uedge = TRUE
@#else
  else if (index(datafilename, '.u') > 0) then
    uedgefilename = datafilename
    uedge = TRUE
  else
    assert('File_name_must_end_in_.d_or_.u' ≡ '.u')
@#endif
  end if

  call read_geometry

  call nc_read_elements
  call nc_read_species
  call nc_read_materials
  call nc_read_reactions
  call nc_read_pmi
  call nc_read_problem

  call setup_back_arrays(geom_modified)

  so_grps = 0
  so_seg_tot = 0
  so_set_run_flags

```

```

so_time_dependent = FALSE

so_gparams_list_size = 0
so_gparams_list_dim = 1
so_params_list_size = 0
so_params_list_dim = 1
so_params_data_size = 0
so_params_data_dim = 1
so_giparams_list_size = 0
so_giparams_list_dim = 1
so_iparams_list_size = 0
so_iparams_list_dim = 1
so_iparams_data_size = 0
so_iparams_data_dim = 1
var_alloc(source_gparameters_list)
var_alloc(source_parameters_list)
var_alloc(source_gparameters_data)
var_alloc(source_parameters_data)
var_alloc(source_giparameters_list)
var_alloc(source_iparameters_list)
var_alloc(source_giparameters_data)
var_alloc(source_iparameters_data)

update = FALSE
if (degas  $\equiv$  TRUE) then
    call read_degas_background(degasfilename)
elseif (uedge  $\equiv$  TRUE) then
    call read_uedge_background(uedgefilename, update)
endif /* Mindlessly copied this section from defineback following an allocation assertion in
    set_background_sources. Note that we do var_reallocb here so that set_background_sources can
    do an explicit var_realloc once the final so_grps is known. The other (segment-based) arrays,
    however, will still be growing one segment at a time. */
if (update  $\equiv$  FALSE) then
    var_reallocb(source_base_ptr)
    var_reallocb(source_num_segments)
    var_reallocb(source_type)
    var_reallocb(source_geometry)
    var_reallocb(source_num_flights)
    var_reallocb(source_num_checkpoints)
    var_reallocb(source_species)
    var_reallocb(source_root_species)
    var_reallocb(source_time_variation)
    var_reallocb(source_num_gparameters)
    var_reallocb(source_num_parameters)
    var_reallocb(source_gparameters_base)
    var_reallocb(source_parameters_base)
    var_reallocb(source_parameters_data_base)
    var_reallocb(source_num_giparameters)
    var_reallocb(source_num_iparameters)
    var_reallocb(source_giparameters_base)
    var_reallocb(source_iparameters_base)
    var_reallocb(source_iparameters_data_base)
    var_reallocb(source_total_current)

```

```
        var_reallocb(source_weight_norm)
        var_reallocb(source_scale_factor)
    end if
    call add_cramd_data    // So we'll have recomb. rates
    call set_background_sources
    call init_wt_alias(uedge)    // Do likewise in boxgen?
    call write_background    /* Possibly relabeled vacuum zones. */
    if (geom_modified  $\equiv$  TRUE)
        call write_geometry
    return
end
```

Set up the background data using a DEGAS input file.

```
"readbackground.f" 1.4 ≡
  @m increment_params(num_params) so_params_list_size += num_params
  if (so_params_list_size > so_params_list_dim) then
    var_realloc(source_parameters_list, so_params_list_dim, so_params_list_size)
    so_params_list_dim = so_params_list_size
  end if

⟨ Functions and Subroutines 1.2 ⟩ +=
  subroutine read_degas_background(file)

    implicit_none_f77
    zn_common
    pr_common
    bk_common
    sp_common
    el_common
    rc_common
    so_common
    gi_common
    sc_common
    implicit_none_f90

    st_decls
    ⟨ Memory allocation interface 0 ⟩

    integer zone, jh, jv, js, jsp, jwall, jseg, seg, iparam     // Local
    real xmin, xmax, zmin, zmax, curr_tot
    character*FILELEN file
    logical new_group

    sp_decl(sp_hyd3)
    sp_decl(degas_species9)
    sc_decl(sector)
    vc_decl(x1)
    vc_decl(x2)

    external lookup_sector, neutralize_species
    integer lookup_sector, neutralize_species

    ⟨ pardef.h 0 ⟩
    ⟨ combal.h 0 ⟩
    ⟨ comst.h 0 ⟩
    ⟨ comgeo.h 0 ⟩
    ⟨ compls.h 0 ⟩
    ⟨ comflg.h 0 ⟩
    ⟨ comrfl.h 0 ⟩
    ⟨ comrat.h 0 ⟩
    ⟨ comsv.h 0 ⟩
    ⟨ comstat.h 0 ⟩
    ⟨ compar.h 0 ⟩

    open (unit = diskin, file = file, status = 'old', form = 'formatted')

    call inpt(diskin, geometry-symmetry, xmin, xmax, zmin, zmax)
    close (unit = diskin)
```

```

sp_hyd_1 = sp_lookup('H')
sp_hyd_2 = sp_lookup('D')
sp_hyd_3 = sp_lookup('T')
nohyds = 0
do js = 1, nphyd
  if (nhyd_js ≠ 0)
    nohyds = nohyds + 1
end do
if (nohyds ≡ 1) then
  degas_species_1 = sp_hyd_nhyd_1
  degas_species_2 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_1)) || '2')
else if (nohyds ≡ 2) then
  degas_species_1 = sp_hyd_nhyd_1
  degas_species_2 = sp_hyd_nhyd_2
  degas_species_3 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_1)) || '2')
  degas_species_4 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_1)) || trim(sp_sy(sp_hyd_nhyd_2)))
  if (degas_species_4 ≡ 0) then
    degas_species_4 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_2)) || trim(sp_sy(sp_hyd_nhyd_1)))
  end if
  degas_species_5 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_2)) || '2')
else if (nohyds ≡ 3) then
  degas_species_1 = sp_hyd_nhyd_1
  degas_species_2 = sp_hyd_nhyd_2
  degas_species_3 = sp_hyd_nhyd_3
  degas_species_4 = sp_lookup(sp_sy(sp_hyd_nhyd_1) || '2')
  degas_species_5 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_1)) || trim(sp_sy(sp_hyd_nhyd_2)))
  if (degas_species_5 ≡ 0) then
    degas_species_5 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_2)) || trim(sp_sy(sp_hyd_nhyd_1)))
  end if
  degas_species_6 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_2)) || '2')
  degas_species_7 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_1)) || trim(sp_sy(sp_hyd_nhyd_3)))
  if (degas_species_7 ≡ 0) then
    degas_species_7 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_3)) || trim(sp_sy(sp_hyd_nhyd_1)))
  end if
  degas_species_8 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_2)) || trim(sp_sy(sp_hyd_nhyd_3)))
  if (degas_species_8 ≡ 0) then
    degas_species_8 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_3)) || trim(sp_sy(sp_hyd_nhyd_2)))
  end if
  degas_species_9 = sp_lookup(trim(sp_sy(sp_hyd_nhyd_3)) || '2')
end if
assert(npis ≤ 9)    // Assume no impurity species !

assert(sp_sy(pr_background(1)) ≡ 'e')
do zone = 1, zn_num
  if (zn_type(zone) ≡ zn_plasma) then
    jh = zn_index(zone, 1)
    jv = zn_index(zone, 2)
    /* Check that both jh and jv are valid. If not, could set some default */
    assert(jh * jv ≠ 0)
    bk_n(1, zone) = denehvt_jh, jv, 1
    bk_temp(1, zone) = electron_charge * tehvt_jh, jv, 1
    vc_set(bk_v(1, zone), zero, zero, zero)
  end if
end do

```

```

do js = 1, nohyds
  /* This assumes only singly charged ions. Same assumption made below. */
  jsp = pr_background_lookup(sp_lookup(trim(sp_sy(degas_species_js)) || '+'))
  assert(jsp > 1)
  bk_n(jsp, zone) = denihvt_jh, jv, 1, js
  bk_temp(jsp, zone) = electron_charge * tihvt_jh, jv, 1, js
  vc_set(bk_v(jsp, zone), zero, zero, zero)
end do
end if
end do /* DEGAS only uses cartesian flow velocities */
background_coords = plasma_coords_cartesian

do js = 1, npis
  do jwall = 1, nowals
    new_group = T
    curr_tot = zero
    do jseg = 1, nosegsxz_jwall
      curr_tot += abs(currxz_jseg, 1, jwall, js)
    end do
    if (curr_tot > zero) then
      do jseg = 1, nosegsxz_jwall
        if (new_group) then
          so_grps++
          var_realloca(source_base_ptr)
          var_realloca(source_num_segments)
          var_realloca(source_type)
          var_realloca(source_geometry)
          var_realloca(source_num_flights)
          var_realloca(source_num_checkpoints)
          var_realloca(source_species)
          var_realloca(source_root_species)
          var_realloca(source_time_variation)

          var_realloca(source_num_gparameters)
          var_realloca(source_num_parameters)
          var_realloca(source_gparameters_base)
          var_realloca(source_parameters_base)
          var_realloca(source_parameters_data_base)
          var_realloca(source_num_giparameters)
          var_realloca(source_num_iparameters)
          var_realloca(source_giparameters_base)
          var_realloca(source_iparameters_base)
          var_realloca(source_iparameters_data_base)

          var_realloca(source_total_current)
          var_realloca(source_weight_norm)
          var_realloca(source_scale_factor)
          so_base(so_grps) = so_seg_tot + 1
          so_nseg(so_grps) = nosegsxz_jwall
          so_nflights(so_grps) = 100
          so_chkpt(so_grps) = 0
          so_t_varn(so_grps) = so_delta_fn

          source_num_gparameters_so_grps = 0
          source_num_parameters_so_grps = 0
        end if
      end do
    end if
  end do
end do

```

```

source_gparameters_baseso-grps = so_gparams_list_size
source_parameters_baseso-grps = so_params_list_size
source_parameters_data_baseso-grps = so_params_data_size
source_num_gparametersso-grps = 0
source_num_iparametersso-grps = 0
source_gparameters_baseso-grps = so_giparams_list_size
source_iparameters_baseso-grps = so_iparams_list_size
source_iparameters_data_baseso-grps = so_iparams_data_size
/* Assume these are same for all segments: */
if (currxtjseg, 1, jwall, js > zero) then
  so_type(so-grps) = so_plate
  so_root_sp(so-grps) = sp_lookup(trim(sp_sy(degas_speciesjs)) || '+' )
  so_species(so-grps) = neutralize_species(so_root_sp(so-grps))
  source_num_parametersso-grps = 3 // Placeholder only
  increment_params(source_num_parametersso-grps)
  so_params_list(1, so-grps) = so_param_e_ion_delta
  so_params_list(2, so-grps) = so_param_e_ion_mult
  so_params_list(3, so-grps) = so_param_e_ion_sheath
else
  so_type(so-grps) = so_puff
  so_species(so-grps) = degas_speciesjs
  so_root_sp(so-grps) = degas_speciesjs
  source_num_gparametersso-grps = 2
  so_gparams_list_size += source_num_gparametersso-grps
  if (so_gparams_list_size > so_gparams_list_dim) then
    var_realloc(source_gparameters_list, so_gparams_list_dim, so_gparams_list_size)
    var_realloc(source_gparameters_data, so_gparams_list_dim, so_gparams_list_size)
    so_gparams_list_dim = so_gparams_list_size
  end if
  so_gparams_list(1, so-grps) = so_gparam_puff_temp
  so_gparams_data(1, so-grps) = t0puffjs * electron_charge
  so_gparams_list(2, so-grps) = so_gparam_puff_exponent
  so_gparams_data(2, so-grps) = one // Hardwired !
end if
  assert(so_species(so-grps) > 0)
  so_geom(so-grps) = so_surface
  so_tot_curr(so-grps) = zero
  so_scale(so-grps) = one // Default
  seg = so_seg_tot
  so_seg_tot += nosegsxzjwall
  var_realloc(source_current, seg, so_seg_tot)
  var_realloc(source_segment_ptr, seg, so_seg_tot)
  var_realloc(source_segment_rel_wt, seg, so_seg_tot)
  var_realloc(source_segment_prob_alias, seg, so_seg_tot)
  var_realloc(source_segment_ptr_alias, seg, so_seg_tot)
  new_group =  $\mathcal{F}$ 
end if /* Any reason not to exclude segments with zero current? */
if ((xwall(jseg, jwall)  $\neq$  xwall(jseg + 1, jwall))  $\vee$  zwall(jseg, jwall)  $\neq$  zwall(jseg + 1,
  jwall))  $\wedge$  currxtjseg, 1, jwall, js  $\neq$  zero) then
  seg++
  source_currentseg = abs(currxtjseg, 1, jwall, js)
  so_tot_curr(so-grps) += source_currentseg

```

```

vc_set(x1, xwall(jseg, jwall), zero, zwall(jseg, jwall))
vc_set(x2, xwall(jseg + 1, jwall), zero, zwall(jseg + 1, jwall))
sector = lookup_sector(x2, x1)
assert(sc_check(sector))
if (so_type(so_grps)  $\equiv$  so_plate) then
    assert(source_num_parametersso_grps > 0)    // Should be 3 !
    so_params_data_size += source_num_parametersso_grps
    if (so_params_data_size > so_params_data_dim) then
        var_realloc(source_parameters_data, so_params_data_dim, so_params_data_size)
        so_params_data_dim = so_params_data_size
    end if    /* Set incident ion description to be  $3T_e$  (sheath potential) plus  $1 \times E_{\text{ion}}$ 
              with  $E_{\text{ion}}$  sampled from a Maxwellian. */
    do iparam = 1, source_num_parametersso_grps
        if (so_params_list(iparam, so_grps)  $\equiv$  so_param_e_ion_delta) then
            so_params_data(iparam, seg, so_grps) = const(3.) * bk_temp(1, sector_zonesector)
        else if (so_params_list(iparam, so_grps)  $\equiv$  so_param_e_ion_sheath) then
            so_params_data(iparam, seg, so_grps) = zero
        else if (so_params_list(iparam, so_grps)  $\equiv$  so_param_e_ion_mult) then
            so_params_data(iparam, seg, so_grps) = one
        end if
    end do
    end if
    source_segment_ptrseg = sector
else
    so_nseg(so_grps) --
    so_seg_tot --
end if
end do
end if
end do
end do
    return
end

```

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⟨ Functions and Subroutines 1.2, 1.3, 1.4 ⟩ Used in section 1.1.
⟨ Memory allocation interface 0 ⟩ Used in sections 1.4 and 1.3.
⟨ combal.h 0 ⟩ Used in section 1.4.
⟨ comflg.h 0 ⟩ Used in section 1.4.
⟨ comgeo.h 0 ⟩ Used in section 1.4.
⟨ compar.h 0 ⟩ Used in section 1.4.
⟨ compls.h 0 ⟩ Used in section 1.4.
⟨ comrat.h 0 ⟩ Used in section 1.4.
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⟨ comstat.h 0 ⟩ Used in section 1.4.
⟨ comsv.h 0 ⟩ Used in section 1.4.
⟨ pardef.h 0 ⟩ Used in section 1.4.

COMMAND LINE: "fweave -f -i! -W[-ybs15000 -ykw800 -ytw40000 -j -n/
/Users/dstotler/degas2/src/readbackground.web".

WEB FILE: "/Users/dstotler/degas2/src/readbackground.web".

CHANGE FILE: (none).

GLOBAL LANGUAGE: FORTRAN.